

DSAA3073: Theories in Data Science

Week 2 Validation Sheet · From Shape to Feasible Learning

Coverage: Week 2

Duration: 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. Which set is convex?

- (a) $\{(-1, 0), (1, 0)\}$
- (b) $\{x \in \mathbb{R}^2 : \|x\|_2 = 1\}$
- (c) $\{x \in \mathbb{R}^2 : x_1 + x_2 = 1\}$
- (d) $\{x \in \mathbb{R}^2 : 1 \leq \|x\|_2 \leq 2\}$

A2 [5 marks]. A differentiable convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies $\nabla f(w^*) = 0$. Which conclusion is guaranteed?

- (a) w^* is a global minimizer, but it need not be unique.
- (b) w^* is the unique global minimizer.
- (c) f must be constant.
- (d) The gradient method converges to w^* for every positive step size.

A3 [5 marks]. What is justified immediately after computing $w_{t+1} = w_t - \eta \nabla f(w_t)$ for a declared $\eta > 0$?

- (a) The update has been executed.
- (b) The objective value has decreased.
- (c) The iterates converge to a unique minimizer.
- (d) The algorithm is polynomial time.

A4 [5 marks]. A program evaluates all 2^d binary candidates. Which conclusion follows from this fact alone?

- (a) The stated program is not polynomial time in variable d .
- (b) No polynomial-time algorithm exists for the problem.
- (c) The problem has a computational–statistical gap.
- (d) The sample size must also be exponential in d .

Part B: Definition in use

B1 [20 marks]. Define

$$C = \{x \in \mathbb{R}^2 : |x_1| + 2|x_2| \leq 4 \text{ and } x_1 - x_2 = 1\}.$$

Decide whether C is convex. Give a proof from the definition using arbitrary points of C and an arbitrary interpolation parameter; a diagram or a few numerical checks is not sufficient.

Answer space

Part C: Basic skills

C1 [7 marks]. Let $f(t) = t^2 - 2t$. For $x = 0$, $y = 3$, and $\lambda = \frac{1}{3}$, compute both sides of the convex-function chord inequality. Does this one chord satisfy the inequality, and what does that single calculation not prove?

Answer space

C2 [8 marks]. For integer $N \geq 2$, a routine has runtime

$$T(N) = N^2 + N2^{\lceil \log_2 N \rceil}.$$

Give a polynomial big- O bound in the encoded input length N and justify it using an explicit bound on $2^{\lceil \log_2 N \rceil}$.

Answer space

Part D: Construction and counterexample

D1 [17 marks]. Construct a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ for which $f'(0) = 0$ but 0 is not a global minimizer. Verify both claims, then give a specific violated chord inequality showing that your example is not convex.

Answer space

D2 [18 marks]. A student claims that the pointwise minimum of two convex functions must be convex. Construct two convex functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ for which $h(x) = \min(f(x), g(x))$ is not convex. Verify that both inputs are convex and give a complete chord witness against h .

Answer space

Part E: A two-sided convexity audit

E1 [25 marks]. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. Suppose that both f and the function $r(x) = -f(x)$ are convex on $\mathbb{R}$. Characterize exactly all functions f with this property and prove both directions of your characterization. A list of examples without a general proof is not sufficient.

Answer space

Part F: Encoding changes the runtime verdict

F1 [15 marks]. The same table-search routine is run under two input formats.

- In format E, the file explicitly lists one one-bit score for each of the 2^d binary strings, in a fixed order. Thus its encoded length is $N_E = \Theta(2^d)$, and the routine scans every listed score once.
- In format S, the file is a circuit description of length $N_S = \Theta(d^3)$. It computes the score of any requested string in time polynomial in d , and the routine requests all 2^d strings.

Classify the stated routine as polynomial or non-polynomial in the encoded input length under each format. Justify both classifications and explain why the phrase “it loops 2^d times” is not by itself an encoding-independent runtime verdict.

Answer space